

PROBLEM

A motorway speed limit, in physics units

The UK motorway speed limit is 70 mph. Every physics equation you will use needs speed in metres per second. Convert it. Take 1 mile = 1609 m.

WHAT YOU KNOW

speed = 70 mph · 1 mile = 1609 m · 1 hour =
3600 s

WHAT YOU WANT

the same speed in m/s

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

A motorway speed limit, in physics units

1 Turn the miles into metres

In one hour the car covers 70 miles.

$$70 \times 1609 \text{ m} = 112\,630 \text{ m}$$

2 Turn the hour into seconds

A speed is metres divided by seconds, so the hour has to go.

$$112\,630 \text{ m} \div 3600 \text{ s} = 31.29 \text{ m/s}$$

3 Quote it sensibly

The speed limit was given to 2 s.f., so 3 s.f. is as far as the data honestly reaches.

$$v = 31.3 \text{ m/s}$$

ANSWER

31.3 m/s

70 mph is just over 31 metres every second.

⚠ THE TRAP – most students answer 19.4 m/s

Dividing by 3.6. That converts km/h to m/s, not mph. Applied to miles per hour it throws away 38% of the speed. The “÷ 3.6” shortcut only becomes valid once the number is already in km/h.

DOES IT MAKE SENSE?

70 mph is about 112.6 km/h, and $112.6 \div 3.6 = 31.29 \text{ m/s}$. Two different routes, the same answer.

PROBLEM

Density of an object you cannot measure with a ruler

A stainless steel nut has a mass of 56.0 g. Dropped into a measuring cylinder, the water level rises from 50.0 cm³ to 57.0 cm³. Find its density in kg/m³.

WHAT YOU KNOW

mass = 56.0 g · water before = 50.0 cm³ · water
after = 57.0 cm³

WHAT YOU WANT

the density in kg/m³

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Density of an object you cannot measure with a ruler

1 The volume is the rise, not the final reading

The nut pushes aside its own volume of water, so the volume you want is the difference.

$$V = 57.0 - 50.0 = 7.0 \text{ cm}^3$$

2 Density = mass ÷ volume

Work in the units you were given, then convert once, at the end.

$$\rho = 56.0 \div 7.0 = 8.0 \text{ g/cm}^3$$

3 Convert to SI

1 g = 1/1000 kg and 1 cm³ = 1/1 000 000 m³, so the two conversions do not cancel: they leave a factor of 1000 upwards.

$$\rho = 8.0 \times 1000 = 8000 \text{ kg/m}^3$$

ANSWER

8000 kg/m³

8.0 g/cm³, which is 8000 kg/m³.

⚠ THE TRAP – most students answer 8.0 kg/m³

Keeping the g/cm³ number and simply changing the label. The answer is then out by a factor of 1000 – and it should look absurd on sight, because 8.0 kg/m³ is lighter than air.

DOES IT MAKE SENSE?

Stainless steel runs about 7900-8100 kg/m³. 8000 sits inside that range, so the answer describes a material that actually exists. That is the fastest check there is on a density.

PROBLEM

Two tugs, one barge

Two tugs pull a barge. One pulls with 8.0 kN due north, the other with 6.0 kN due east. Find the magnitude and direction of the resultant force.

WHAT YOU KNOW

$F_{\text{north}} = 8.0 \text{ kN}$ · $F_{\text{east}} = 6.0 \text{ kN}$ · angle
between them = 90°

WHAT YOU WANT

the resultant force: magnitude and direction

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Two tugs, one barge

1 Draw the two forces tip to tail

They are at 90° to each other, so the two forces and the resultant form a right-angled triangle.

the resultant is the hypotenuse

2 Pythagoras gives the magnitude

This is the whole point of calling force a vector: you cannot simply add the numbers.

$$F = \sqrt{(8.0^2 + 6.0^2)} = \sqrt{100} = 10.0 \text{ kN}$$

3 Trigonometry gives the direction

Measure it from north, the way a bearing is always quoted.

$$\theta = \tan^{-1}(6.0 \div 8.0) = 36.9^\circ \text{ east of north}$$

ANSWER

10.0 kN at 36.9° east of north

It is a 3-4-5 triangle scaled up by two, which is why the numbers come out clean.

⚠ THE TRAP – most students answer 14 kN

Adding the magnitudes. That is only correct when two forces point the same way. Here it overstates the pull by 40% – and it produces no direction at all, which is the clue that the method itself is wrong.

DOES IT MAKE SENSE?

A resultant must always lie between the difference of the two forces (2.0 kN) and their sum (14.0 kN). 10.0 kN sits inside that window.

PROBLEM

Reading a speed off a distance-time graph

A distance-time graph is a straight line. At $t = 2.0$ s the distance is 14.0 m; at $t = 6.0$ s it is 34.0 m. Find the speed.

WHAT YOU KNOW

point A = (2.0 s, 14.0 m) · point B = (6.0 s, 34.0 m)

WHAT YOU WANT

the speed of the object

YOUR WORKING

Give it five minutes before you turn the page.

SOLUTION

Reading a speed off a distance-time graph

1 Speed is the gradient, not the height

On a distance-time graph the gradient is distance per unit time, which is exactly what speed means.

$$v = \Delta d \div \Delta t$$

2 Take the differences

Use two points that are far apart; it keeps any reading error small.

$$\Delta d = 34.0 - 14.0 = 20.0 \text{ m and } \Delta t = 6.0 - 2.0 = 4.0 \text{ s}$$

3 Divide

$$v = 20.0 \div 4.0 = 5.0 \text{ m/s}$$

ANSWER

5.0 m/s

Constant speed, because the line is straight.

⚠ THE TRAP – most students answer 5.67 m/s

Taking a single point and dividing distance by time ($34.0 \div 6.0$). That only works if the line passes through the origin. This one does not – it starts at 4.0 m – so the single-point method comes out 13% too high.

DOES IT MAKE SENSE?

Test it on the other point: the line predicts $4.0 + 5.0 \times 2.0 = 14.0 \text{ m}$ at $t = 2.0 \text{ s}$, which is what the graph shows.